Peter Laszcz

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I am currently in my final year of an Integrated Masters in Mathematics and Computer Science at the University of Birmingham. I have developed skills in software development and mathematical modelling through both employment and educational experience. I supplement my academic interests through personal and independent research and projects, and I wish to pursue a career in research with computer science and applied mathematics after I graduate.

Work Experience

Scientific Software Engineer Intern with Science and Technology Facilities Council (STFC) **(May-Sept 2026)**

- Software development position at the ISIS Neutron and Muon Source, Rutherford Appleton Laboratory.
- Developing data analysis, processing, and visualisation software for a novel neutron imaging method.
- Collaborating internally to integrate the code within the facility's long-term processing codebase, and externally on an international level to bring the program to other neutron imaging facilities.
- Documenting the developments encapsulated within the program in a paper to be submitted for first-author publication.
- Received training in CI/CD, project management, and documentation systems.
- Gave presentations and talks on the project, as well as producing outreach content for the facility's social media.

Open-Source Developer with Google Summer of Code **(May-Sept 2025)**

- Stipend received from Google via the Python Software Foundation to develop features of Ilastik (image-processing software) under the mentorship of Research Software Engineers at the European Molecular Biology Laboratory (EMBL), Heidelberg.
- Operated within an international group with Agile methodology and Test-Driven Development principles.
- Delivered project progress updates at weekly team meetings.
- Merged a highly-requested feature (physical pixel sizes) by successful completion of the project.

Research Associate at the University of Birmingham **(Oct-Dec 2024)**

- Programmer and mathematical modeller for ARME, an EPSRC-funded project investigating musician synchronisation in ensemble performance.
- Responsibilities included: researching novel synchronisation model literature; implementing models in multiple languages (Python, MATLAB, C++); performing model analysis with experimental data; managing the source code via Version Control Systems; providing code and model documentation; attending weekly team meetings to present updates and results.
- Findings resulted in paper co-authorship (currently in review, PLOS Computational Biology)

Research Intern at the University of Birmingham Positron Imaging Centre **(Jun-Aug 2024)**

- Conducted experimental data analysis on reactor systems in the context of plastics pyrolysis.
- Utilised particle-tracking data to model fluid dynamics within the system, as well as analysing radiographs via Convolutional Neural Networks to locate and quantitively measure gas structures.
- Deployed and maintained novel software on the university's HPC system, as well as developing features of existing in-house software.
- Discussed the aims, progress, and impact of the research project with academics from the university and abroad at conferences.
- Yielded a 54% improvement in structure measurement over classical techniques, as well as enabling the analysis of additional bed structures.

Education

M.Sci Mathematics and Computer Science at the University of Birmingham **(Sept 2023 – Jun 2027)**

Final year, graduating 2027. Predicted First Class Honours.

A-Levels at the Cardinal Vaughan Memorial School **(Sept 2021 – Jun 2023)**
Mathematics A*, Physics A*, Chemistry A, Further Mathematics B

GCSEs at St. Joan of Arc Catholic School, Rickmansworth **(Sept 2016 – Jun 2021)**
Achieved grades 9-7 (seven 9s, two 8s, one 7). Outstanding Achievement award in Mathematics and Computer Science.

Volunteering Positions

Secretary at University of Birmingham Disabled Students' Association (SANDAM) **(2025-2026)**

- Minuting meetings and processing administration data.
- Communicating with student union and university administration, and collaborating with faculties, to raise awareness and promote accessibility for disabled students.
- Booking rooms to ensure meetings and events can go ahead.

Overseas Mobility Assistant with Diocese of Westminster Redcaps **(2024)**

- Worked in a team to provide wheelchair aid to mobility-assisted people in the French Pyrenees.
- Paired with a volunteer throughout, resulting in developed skills in one-to-one communication and fostering working relationships, as well as interpersonal skills with our mobility-assisted client.
- The work raised my awareness of social and infrastructure issues faced by mobility-assisted individuals.

Treasurer at Ruislip SVP and University of Birmingham G&S Society **(2021-2022; 2023-2026)**

- Collecting & processing financial and administration data, thereby strengthening proficiency in organisation.
- Collating receipts and invoices to produce and present reports and determine methods of managing costs.
- Working with event organisers to calculate budgets and apply for grants.

Young Scout Leader at 4th Eastcote Scout Troop **(2021-2022)**

- Planned and organised a weekly activity schedule with a team of leaders for a scout patrol, building confidence in project management.
- Ran schedule with patrol, bolstering skills in leadership and communication with large groups, as well as teamwork alongside my fellow Scout Leaders.

Non-Academic Achievements and Interests

- **Programming:** Proficiency in multiple programming (C, C++, Haskell, Java, Python, MATLAB) and markup (LaTeX, Markdown) languages. Project interests include Computer Vision, cryptography, and High Performance Computing. Experience in hackathons (BirmingHack 2025; BirmingHack 2026 (second place); University of Birmingham Secure Compute Hackathon 2026 (produced whitepaper on implementation of Trusted Research Environments in High-Performance Computing for university executives)).
- **Institute of Mathematics and its Applications (IMA):** Associate Member.
- **Mathematics Engagement:** Previous engagement work has included mathematics aid at Peer Assisted Study Sessions, and secondary-level demonstrations with the IMA at a national science and technology fair.
- **University Societies:** Gilbert & Sullivan Society, Music Society, Maths Society, Computer Science Society, Student Disability Association.
- **Music:** Violin Grade 8, 13 years choral experience (recent performances include a collaboration with the BBC Singers, and at St. Paul's Cathedral, London), and 10 years orchestral experience (most recently performing in a tour in Sicily).
- **Dance:** Phoenix Morris Dancer and Fiddle Player; 8 years experience.
- **Scouting:** Member of REN Network Group; Gold Duke of Edinburgh Award holder.

Reference

Dr. Galane J. Luo (Personal Academic Tutor)

Lecturer in the School of Mathematics at the University of Birmingham

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